UNIT 5 — PHYSIOLOGY

Comprehensive revision notes based on the supplied textbook pages, with key explanations, comparisons and useful scientific clarifications.

PART A — PLANT PHYSIOLOGY

1. PHOTOSYNTHESIS

Autotrophs are organisms that prepare their own food from simple substances. Green plants are producers because they make organic food using light energy.

• **Chlorophyll** is the green pigment in chloroplasts that absorbs light energy. Light energy is converted into chemical energy stored in carbohydrates.

• **Photosynthesis** is the process by which green plants use light energy to make carbohydrates from carbon dioxide and water, releasing oxygen.

Overall simplified equation: 6CO■ + 6H■O —(light, chlorophyll)→ C■H■■O■ + 6O■

• Carbon dioxide enters mainly through stomata; water is absorbed by roots and transported to leaves through xylem.

• Oxygen released during photosynthesis comes from the splitting of water molecules in the light-dependent reactions.

• Photosynthesis requires light, chlorophyll, carbon dioxide and water. Temperature also affects the rate because enzymes are involved.

Starch test in a leaf

• Boil the leaf briefly in water, then heat it in ethanol using a water bath to remove chlorophyll. Rinse/soften it and add iodine solution.

• A blue-black colour indicates the presence of starch, showing that photosynthesis has produced stored carbohydrate.

2. TRANSLOCATION

Translocation is the transport of soluble products of photosynthesis, especially sucrose, from leaves (sources) to parts of the plant where they are used or stored (sinks), such as roots, fruits, growing shoots and seeds.

• Translocation occurs through **phloem**. Phloem can transport food both upward and downward according to the plant's needs.

• Do not confuse translocation with water transport: **xylem** mainly carries water and dissolved mineral salts from roots toward the aerial parts.

3. TRANSPIRATION

Transpiration is the loss of water vapour from the aerial parts of a plant, mainly through leaves. Water evaporates from moist cell surfaces and diffuses out through openings.

Type	Route
Stomatal	Through stomata; usually the major route in leaves.
Cuticular	Through the waxy cuticle covering the epidermis.
Lenticular	Through lenticels in woody stems.

Importance of transpiration

• Creates a transpiration pull that helps move water and dissolved minerals upward through xylem (ascent of sap).

• Helps cool the plant by evaporation.

• Contributes to the movement and distribution of water and minerals and helps maintain water balance.

• Excessive transpiration can cause wilting if water loss exceeds water absorption.

Factors affecting transpiration

• Higher temperature, stronger air movement and lower humidity generally increase transpiration; high humidity reduces it.

• Stomata can open or close to regulate gas exchange and water loss. Most stomata open in light in many plants, though exceptions exist.

4. RESPIRATION IN PLANTS

Respiration is the cellular process in which food molecules, especially glucose, are broken down to release energy. Cells capture much of this energy in ATP, which powers cellular activities.

Aerobic respiration: glucose + oxygen → carbon dioxide + water + energy (ATP).

• Plants respire continuously, during both day and night. Photosynthesis occurs only when suitable light is available.

• Gas exchange occurs by diffusion: through stomata in leaves and green stems, lenticels in woody stems, and the surface/root hairs in roots.

• In young roots, oxygen from air spaces between soil particles diffuses into root tissues; carbon dioxide diffuses out.

• In herbaceous green stems, stomata allow gas exchange; in woody stems, lenticels perform this role.

Photosynthesis vs respiration

Photosynthesis	Respiration
Stores light energy as chemical energy in food.	Releases usable energy from food and transfers it to ATP.
Uses carbon dioxide and water; produces carbohydrate and oxygen.	Aerobic respiration uses oxygen and produces carbon dioxide and water.
Occurs in chlorophyll-containing cells when light is available.	Occurs in living cells day and night.

5. PLANT MOVEMENTS

Plants respond to stimuli such as light, gravity, water and touch. A directional growth response is called a **tropism**.

Response	Meaning / example
Phototropism	Growth response to light; shoots are usually positively phototropic (grow toward light).
Geotropism / gravitropism	Growth response to gravity; roots are usually positively geotropic, shoots negatively geotropic.
Hydrotropism	Growth response to water; roots tend to grow toward moisture.
Thigmotropism	Growth response to touch; tendrils coil around supports.

PART B — ANIMAL PHYSIOLOGY

6. NUTRITION AND DIGESTION

Nutrition is the intake of food and its use by the body for energy, growth, repair and maintenance. In humans, the main stages are: **ingestion** → **digestion** → **absorption** → **assimilation** → **egestion**.

• **Ingestion:** taking food into the mouth. **Digestion:** breaking complex food into simpler soluble substances.

• **Absorption:** digested nutrients pass through the intestinal wall into blood or lymph. **Assimilation:** cells use absorbed nutrients.

• **Egestion:** removal of undigested material through the anus; it is different from excretion, which removes metabolic wastes.

Human alimentary canal and digestive glands

Part / organ	Main function
Mouth / buccal cavity	Teeth mechanically break food; tongue mixes food and helps swallowing. Saliva moistens food and salivary amylase (ptyalin) begins starch digestion.
Pharynx	Common passage region for food and air; helps swallowing.
Oesophagus	Muscular food pipe; moves food to stomach by peristalsis. No significant digestion occurs here.
Stomach	Muscular sac. Gastric juice contains hydrochloric acid and enzymes such as pepsin; acidic conditions help protein digestion and kill many microbes.
Small intestine	Completes most chemical digestion and absorbs nutrients. Villi and microvilli greatly increase absorptive surface area.
Large intestine	Absorbs water and salts; forms and stores faeces before egestion.
Liver	Produces bile, which helps emulsify fats. Bile is stored in the gall bladder and released into the small intestine.
Pancreas	Releases pancreatic juice containing enzymes such as amylase, lipase and proteases into the small intestine.

Digestive enzymes and important actions

• **Salivary amylase:** begins digestion of starch into smaller sugars.

• **Pepsin:** begins protein digestion in the stomach; it works in acidic conditions.

• **Pancreatic amylase:** digests starch; **trypsin and other proteases** digest proteins; **lipase** digests fats.

• **Bile is not an enzyme.** It breaks large fat globules into smaller droplets (emulsification), increasing the surface area for lipase.

Human teeth

Type	Function
Incisors	Cut and bite food.
Canines	Tear food.
Premolars	Crush and grind food.
Molars	Grind and crush food.

• Milk teeth: 20 in total, usually begin appearing at about 6 months. Permanent teeth: usually 32 in total, including wisdom teeth; eruption begins around 6 years.

• Dental formula for adult human: 2.1.2.3 / 2.1.2.3 (per half of upper and lower jaws). Milk dentition: 2.1.0.2 / 2.1.0.2.

7. RESPIRATION IN ANIMALS

Respiration releases energy from food in cells. Breathing is the physical movement of air into and out of respiratory organs; it is not the same as cellular respiration.

Type	Respiratory surface / organ	Examples
Body-surface diffusion	General body surface	Amoeba, Paramecium (single-celled organisms)
Tracheal	Spiracles and tracheal tubes	Insects such as cockroach
Branchial	Gills	Fish and many aquatic animals
Cutaneous	Moist skin	Earthworm; frogs can also exchange gases through skin
Pulmonary	Lungs	Mammals, adult amphibians, reptiles and birds

8. HUMAN RESPIRATORY SYSTEM

Structure	Function
Nostrils and nasal cavity	Air enters; hairs and mucus trap particles. The nasal cavity warms and moistens incoming air.
Pharynx	Passage shared by the digestive and respiratory systems.
Larynx	Voice box; contains vocal cords.
Trachea	Windpipe supported by C-shaped cartilage rings that help prevent collapse.
Bronchi	Two main branches of the trachea, one entering each lung.
Bronchioles	Smaller branches within lungs.
Alveoli	Tiny thin-walled air sacs surrounded by capillaries; main site of gas exchange.
Lungs and pleura	Lungs contain branching airways and alveoli; pleural membranes cover the lungs.
Diaphragm	Muscular sheet below lungs that changes chest-cavity volume during breathing.

Mechanism of breathing

• **Inhalation:** diaphragm contracts and flattens; external intercostal muscles raise the ribs. Chest volume increases, pressure inside falls and air enters the lungs.

• **Exhalation:** diaphragm and external intercostal muscles relax; chest volume decreases and air moves out. Quiet exhalation is mostly passive.

• Oxygen diffuses from alveolar air into blood; carbon dioxide diffuses from blood into alveoli and is exhaled.

9. NERVOUS SYSTEM

The nervous system detects stimuli, processes information and coordinates responses. Its main parts include the **brain, spinal cord, nerves and sense organs**.

• **Brain:** protected by the skull; interprets sensory information and coordinates many body activities, thinking, learning and memory.

• **Spinal cord:** lies inside the vertebral column; carries messages between brain and body and coordinates many reflex actions.

• **Nerves:** bundles of nerve fibres that carry electrical impulses between the central nervous system and body parts.

• **Reflex action:** a rapid, automatic response to a stimulus. A reflex arc commonly involves a receptor, sensory neuron, spinal cord/interneuron, motor neuron and effector.

10. EXCRETORY SYSTEM

Excretion is removal of metabolic waste products from the body. It is different from egestion, which removes undigested food.

Organ	Waste / function
Kidneys	Filter blood and remove urea, excess salts and water; form urine.
Lungs	Remove carbon dioxide and water vapour during exhalation.
Skin	Sweat glands release sweat containing water, salts and small amounts of urea.

• The main functional unit of the kidney is the **nephron**, which filters blood and helps form urine.

11. BLOOD GROUPS

The ABO blood group system depends on the presence or absence of A and B antigens on red blood cells. The Rh system commonly refers to the D antigen: Rh-positive means D antigen is present; Rh-negative means it is absent.

Group	Red-cell antigens	Plasma antibodies (simplified)
A	A	Anti-B
B	B	Anti-A
AB	A and B	Neither anti-A nor anti-B
O	Neither A nor B	Anti-A and anti-B

• **Red-cell transfusion (ABO/Rh simplified):** O negative is often called the universal red-cell donor; AB positive is often called the universal red-cell recipient. Real transfusions require crossmatching and clinical testing.

Recipient	Compatible red-cell groups (simplified)
O–	O–
O+	O–, O+

Recipient	Compatible red-cell groups (simplified)
A–	O–, A–
A+	O–, O+, A–, A+
B–	O–, B–
B+	O–, O+, B–, B+
AB–	O–, A–, B–, AB–
AB+	All ABO/Rh groups

12. BODY MOVEMENTS AND SKELETAL SYSTEM

Movement is a change in position of a body part. **Locomotion** is movement of the whole organism from one place to another. A skeleton supports the body, protects organs and provides attachment for muscles.

• **Endoskeleton:** internal skeleton, as in humans. **Exoskeleton:** hard external supporting covering, as in a snail shell or tortoise shell.

Human skeleton

• An adult human skeleton typically has **206 bones**. Babies have more separate bones (about 270–300); some fuse during growth.

Part	Structure and function
Skull	Protects the brain; supports the face and sensory structures. The lower jaw (mandible) is movable.
Backbone / vertebral column	Usually 33 vertebrae in early life; some fuse in adults. Protects spinal cord and supports the body.
Rib cage	12 pairs of ribs (24 ribs) protect the heart and lungs and assist breathing.
Limbs	Arms and legs; help in movement and handling objects.
Pelvic girdle	Supports the trunk and protects pelvic organs; attaches lower limbs to the axial skeleton.
Shoulder girdle	Each side has a clavicle (collar bone) and scapula (shoulder blade); attaches upper limb to trunk.

Joints

Joint	Movement / location
Fixed / immovable	Little or no movement; many joints between skull bones.
Hinge	Movement mainly in one plane, like a door; elbow and knee.
Ball-and-socket	Movement in many directions, including rotation; shoulder and hip.
Pivot	Rotation around an axis; example, joint between first and second cervical vertebrae.
Gliding	Bones slide over one another; examples in wrist and ankle.

Muscles and movement

• Skeletal muscles are attached to bones by tendons. Muscles contract and relax to move bones across joints.

• Many skeletal muscles work in antagonistic pairs: when one contracts, the other relaxes. Muscles pull; they do not push bones.

• **Cartilage** is a flexible connective tissue found at the ends of many bones, in the nose and external ear; it reduces friction and cushions joints.

Movement in other animals

• **Earthworm:** circular and longitudinal muscles work with tiny bristles called setae to help movement.

• **Snail:** moves using a muscular foot; mucus reduces friction.

• **Cockroach:** has three pairs of legs and usually two pairs of wings; its exoskeleton is made of chitin.

• **Fish:** streamlined body, muscles, tail fin and other fins help swimming.

• **Snake:** moves using coordinated muscles, ribs and body scales in wave-like patterns.

• **Birds:** wings, feathers, strong flight muscles and lightweight adaptations help flight.

13. QUICK REVISION FACTS

• Photosynthesis stores energy; respiration releases usable energy as ATP.

• Phloem transports sugars; xylem transports water and minerals.

• Transpiration is water-vapour loss; stomatal transpiration is usually the major type in leaves.

• Plants respire in both day and night; photosynthesis requires suitable light.

• Villi increase the absorptive surface area of the small intestine.

• Bile emulsifies fats but is not a digestive enzyme.

• Alveoli are the main gas-exchange surfaces in human lungs.

• The diaphragm contracts during inhalation and relaxes during quiet exhalation.

- Kidneys remove urea; lungs remove carbon dioxide; skin releases sweat.
- Adult human skeleton: 206 bones; 12 pairs of ribs; 32 permanent teeth in a complete adult dentition.
- Hinge joint: elbow/knee; ball-and-socket: shoulder/hip; pivot: neck; gliding: wrist/ankle.

Use the diagrams and tables to revise structures, functions, pathways and comparisons. Learn both definitions and the reasoning behind each process.